**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI**
**First Semester 2019-2020**
**ME/ MF F211: Mechanics of Solids**
**Comprehensive Regular Examination**

**Date: 6.12.19** **Time: 1 Hr** **Max Marks: 30**

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**Note: (1) There are total 4 questions**
**(2) Both side of the paper is printed.**
**(3) The time allotted is 1 hr for Part-A but you can take time as per your convenience. After submitting Part- A, you can take the question paper and answer sheet for Part- B.**

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## PART-A(CLOSED BOOK)

Q1. Bracket *ABCD* of solid circular cross-section has the shape and dimension ($b$ = 37mm) as shown in FigQ1. A vertical and horizontal load $P$ (40 N) acts at the free end $D$.

(a) Draw the bending moment diagram of bracket *ABCD*. **Hint**: Show the variation of bending moment along segments *AB*, *BC* and *CD* separately.

(b) Determine the minimum permissible diameter $d_{min}$ of the bracket, if the allowable bending stress in the material is 30 MPa. (Neglect the weight of bracket) **[8]**

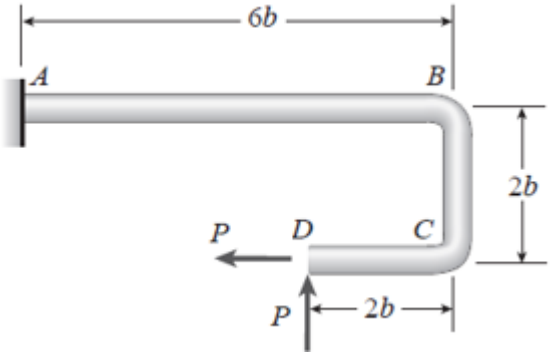



**Fig Q1**

Q2. The principal strains in the plane of a flat aluminum plate loaded in its plane are $\varepsilon_1 = -8.1\times10^{-4}$ $\varepsilon_2 = 5.7\times10^{-4}$. Using Mohr's circle, find $\sigma_x$, $\sigma_y$, and $\tau_{xy}$ where the $x$ and $y$ axes are as shown in FigQ2. Take $E$ = 72GPa and $\nu$ = 0.33. **[7]**

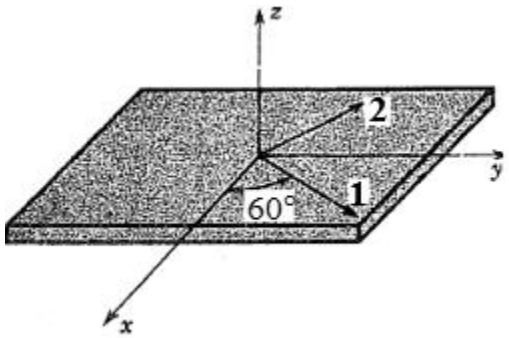



**Fig Q2**

Q3. The curved cantilever beam in FigQ3 has a radius of curvature *R* and a circular cross section of diameter *d*. It is subjected to a point load *P* at point *B*.

(a) Determine the reactions (i.e. axial force, shear force and bending moment) acting at any section θ in the curved cantilever beam *AB*.

(b) Using Castigliano's method (the strain energy method), determine the vertical deflection of point *B* in terms of *P*, modulus of elasticity *E*, radius of curvature *R*, cross-section area *A*, and moment of inertia of the cross section *I*. **Neglect the effect of strain energy due to shear** **[8]**

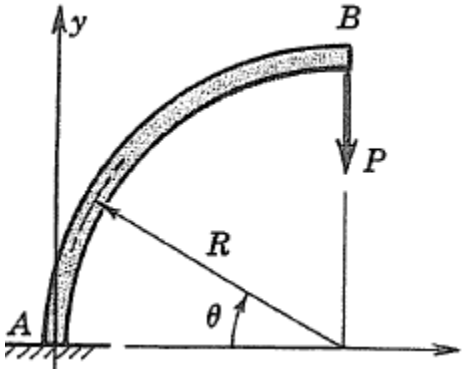



**Fig Q3**

Q4. The state of plane stress shown (in **FigQ4**) occurs at a critical point of a steel machine components. As a result of several tensile tests, it has been found that the tensile yield strength is *Y* = 270 MPa for the grade of steel used. Determine the factor of safety with respect to yield, using

(a) maximum-shearing-stress (MSS) criterion and (b) Distortion energy criterion **[7]**

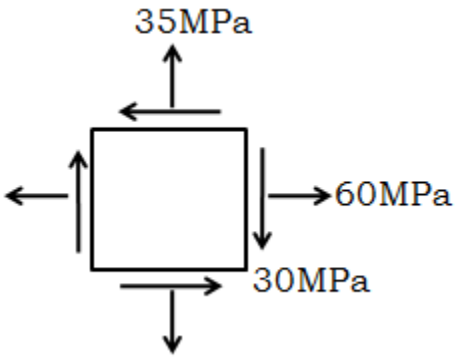



**Fig Q4**

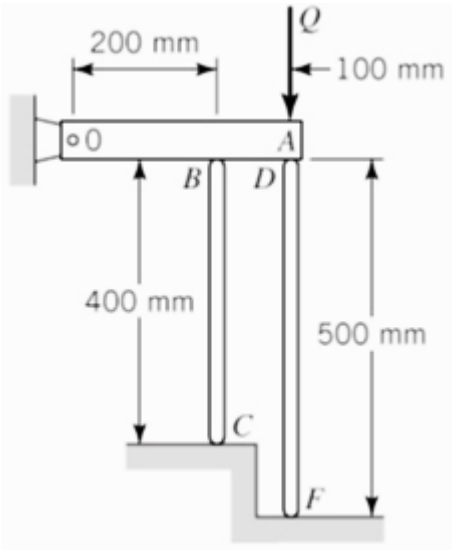
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**Date: 6.12.19** **Time: 2 Hr** **Max Marks: 50**

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**PART-B (OPEN BOOK)**

Q1. In Fig Q1, member *OA* can be considered rigid and weightless. Column *BC* is pin-ended and column *DF* is pin-ended at *A* and fixed at F. Both the columns are of solid circular cross-sections of diameter 10mm. Column BC is made of structural steel ($E$ = 200GPa) and column *DF* is made of an Aluminium alloy ($E$ = 72GPa). Determine the magnitude of $Q$ that will cause one of the columns to buckle. **[10]**




Q2. The box beam is subjected to the 26 kN inclined force (as shown in FigQ2) that is applied at the center of its width, 75 mm from each side.

**a.** Calculate the axial force $F$, shear force $V$, and bending moment $M_b$ acting on the cross section *x-x*.

**b.** Determine the stress components (i.e. $\sigma_x$ and $\tau_{xy}$) on elements at points *A* and *B*.

**c.** Show these stresses on properly oriented 2-D elements.

**d.** Determine the principal stresses (i.e. $\sigma_1$ and $\sigma_2$) at point A. **[18]**

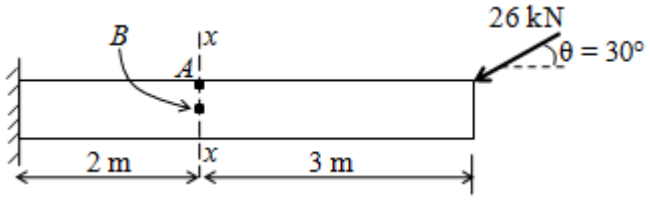
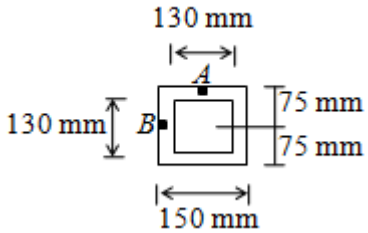



**Fig Q2**

**Section *x-x***:



Q3. The beam *AB* in FigQ3 is fixed supported to the wall at *A* and pin connected to a 12 mm diameter rod *BC* at *B*. The beam is subjected to a point load $P$ = 40 kN at D as shown in Fig. If $E$ = 210 GPa for both the members (i.e. beam *AB* and rod *BC*),

a) Determine the force developed in the rod *BC* due to the loading.

b) State the proper boundary conditions and starting from the fourth-order differential equation (i.e. the load equation), derive the equation of deflection curve for the beam AB.

c) Determine the deflection of point *B* on the beam.

The moment of inertia of the beam about its neutral axis is $I_{NA}$= 186 x $10^6$ $mm^4$. **[10]**

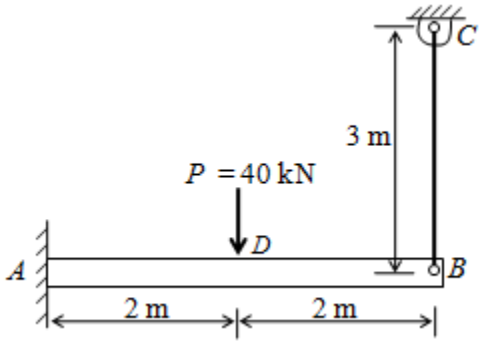



**Fig Q3**

Q4. The 60-mm diameter shaft *ABC* is supported by two journal bearings, while the 80-mm diameter shaft *EH* is fixed at *E* and supported by a journal bearing at *H* (FigQ4). If the angle of twist at gears A and C is required to be 0.05 rad, determine the magnitudes of the torques $T_1$ and $T_2$. The shafts are made of A-36steel ($G$ = 80 GPa). **[12]**

